



Software Process Improvement –SEPI-471



PSP –Personal Software Process & Team Software Process

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Agenda

- PSP-Personal Software Process
- TSP-Team Software Process

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PSP and TSP

- Emphasize the need
 - to continuously collect data about the work that is being performed and to use that data to develop strategies for improvement.
- Watts Humphrey, the developer of both methods.

Watts Humphrey

The PSP [and TSP] will show you **how to plan and track your work and how to consistently produce high quality software.**

Using PSP [and TSP] **will give you the data that show the effectiveness of your work and identify your strengths and weaknesses.**

To have a successful and rewarding career, you need to know your skills and abilities, strive to improve them, and capitalize on your unique talents in the work you do

PSP-Personal software process

- Every developer uses some process to build computer software.
- The process
 - may be haphazard or ad hoc;
 - may change on a daily basis;
 - may not be efficient, effective, or even successful; but a “process” does exist.



Personal Software Process (PSP)

- PSP helps individual software engineers manage their work effectively."
- Key Points:
 - "Guidelines and tools for planning, tracking, and improving software development."
 - "Focuses on personal improvement in coding, testing, and time management"

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PSP-Personal software process

The Personal Software Process (PSP) emphasizes personal measurement of both the **work product** that is produced and the resultant **quality of the work product**.

In addition, PSP makes the practitioner responsible for **project planning (e.g., estimating and scheduling)** and empowers the practitioner to control the quality of all software work products that are developed.



Why is PSP Important?

- Enhances individual productivity.
- Improves software quality.
- Increases predictability in project timelines.

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Components of PSP

- Time Management:
 - Planning time for each task.
 - Tracking time spent on coding, testing, and other activities.
- Quality Management
 - Defining coding standards.
 - Performing thorough testing.
- Process Improvement:
 - Analyzing performance after each project.
 - Identifying areas for personal improvement.

PSP Phases-SD



Planning: Setting goals and estimating tasks.

Design: Outlining the structure of the software.

Code: Writing and testing the code.

Test: Verifying the functionality and fixing bugs.

Postmortem: Reflecting on the process and lessons learned.

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PSP Levels

- **PSP0, PSP0.1 (Introduces process discipline and measurement).**
- PSP0 has 3 phases: **planning, development (design, code, compile, test) and a post mortem.** A baseline is established for current process measuring: time spent on programming, faults injected/removed and size of a program. In a post mortem, the engineer ensures all data for the projects has been properly recorded and analysed.
- PSP0.1 advances the process by adding a **coding standard, a size measurement and the development of a personal process improvement plan (PIP).** In the PIP, the engineer records ideas for improving his own process.

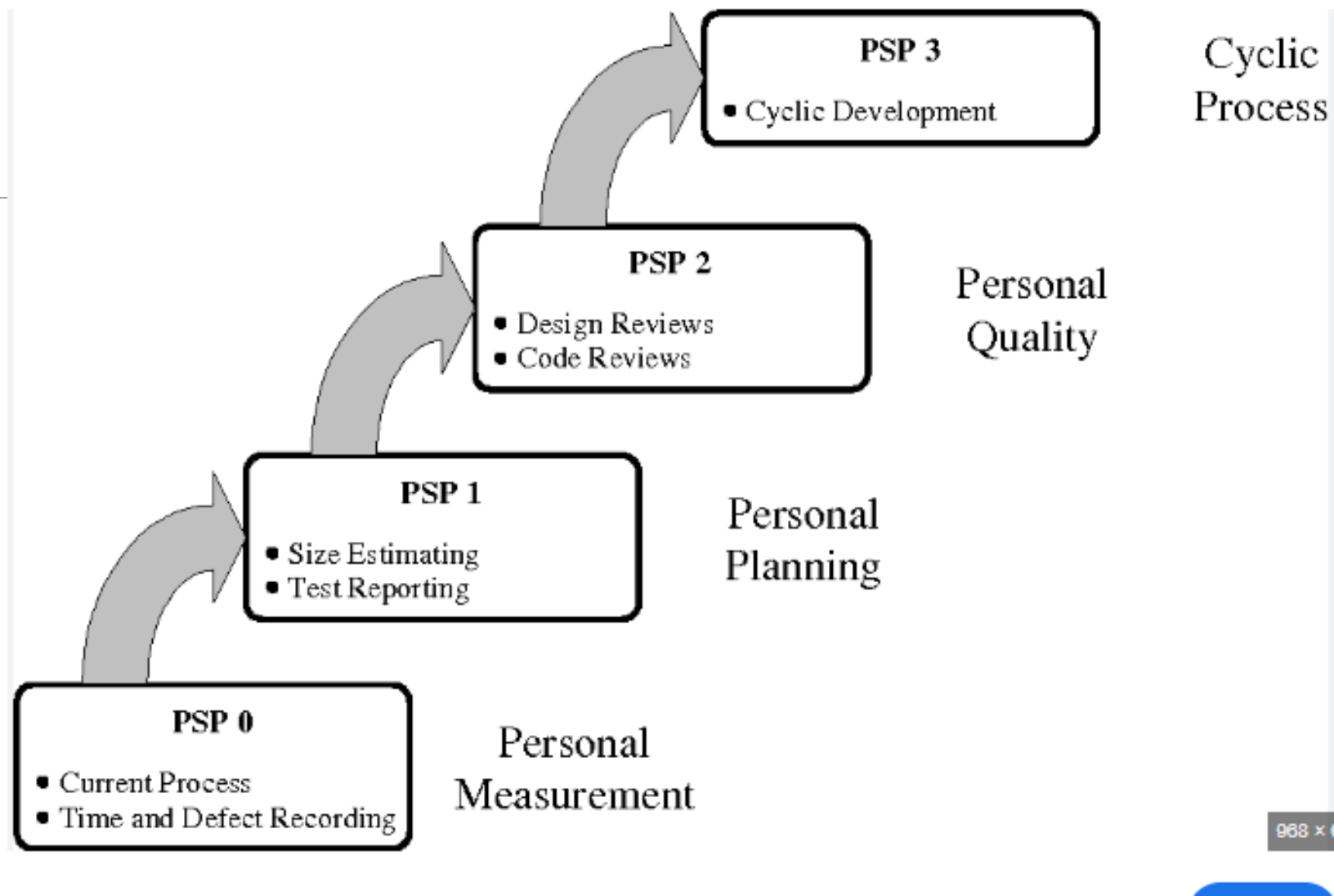
PSP Levels

- PSP1, PSP1.1 (Introduces estimating and planning)
- Based upon the baseline data collected in PSP0 and PSP0.1, the engineer estimates how large a new program will be and prepares a test report (PSP1).
- Accumulated data from previous projects is used to estimate the total time.
- Each new project will record the actual time spent. This information is used for task and schedule planning and estimation (PSP1.1).

PSP Levels

- PSP2, PSP2.1 (Introduces quality management and design)
 - PSP2 adds two new phases: **design review and code review**.
 - **Defect prevention and removal of them are the focus at the PSP2**. Engineers learn to **evaluate and improve their process by measuring how long tasks take and the number of defects they inject** and remove in each phase of development.
 - Engineers **construct and use checklists for design and code reviews**.
 - PSP2.1 introduces design specification and analysis techniques
 - **PSP Level 3 have a thorough understanding of the PSP framework and its various components, including planning, design, coding, testing, and postmortem analysis**. They can effectively manage each phase of the software development lifecycle.

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Certification

- A certification covering PSP is offered by the SEI at Carnegie Mellon University.
- The steps to becoming an SEI-Certified PSP Developer are:
 - learn the PSP;
 - take the certification exam;
- The PSP Developer examination is based on concepts found in the PSP Body of Knowledge.

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PSP

- The PSP is a personal process that can be adapted to suit the needs of the individual developer.
- It is not specific to any programming or design methodology; therefore, it can be used with different methodologies, including Agile software development etc.

TSP

- Team Software Process (TSP) is a structured software development methodology developed by the Software Engineering Institute (SEI) at Carnegie Mellon University. It builds on the Personal Software Process (PSP) and aims to help software development teams improve quality, productivity, and predictability.

TSP -Benefits

- Higher product quality (fewer defects)
- Accurate planning and tracking
- Strong team accountability
- Process repeatability and improvement



TSP- Components

- Team Building- Roles and responsibilities
- Team Work-Processes and practices.

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TSP- When to use

- Mission-critical or safety-critical projects
- Organizations seeking CMMI compliance
- Projects requiring formal process discipline

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Thank you

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